

Physiverse

Version: 1.0

SUMMARY

Build the world's most immersive and interactive physics learning ecosystem where students can experience, experiment, and master physics concepts through 3D simulations and intelligent learning.

DEVELOPMENT PHILOSOPHY

Build in layers.

Do not build everything at once.

Every release should provide value.

PRODUCT LIFECYCLE

Foundation ↓ MVP ↓ Version 1 ↓ Version 2 ↓ Version 3 ↓ Global Platform

PHASE 0 – FOUNDATION

Duration: 2 Weeks

Goals: - Setup architecture - Design system - Authentication - Database - CI/CD

Deliverables: - Next.js setup - Supabase setup - Tailwind setup - R3F setup - Deployment pipeline - Analytics setup

PHASE 1 – MVP

Duration: 6 Weeks

Goal: Launch first public version.

Features: Authentication: Login, Signup, OAuth, Profiles Homepage: Hero section, Physics domains, Simulations showcase, Pricing, Footer Learning System: Subjects, Units, Chapters, Lessons, Progress tracking Simulations: 10 simulations, Interactive controls, Results panel Formula Explorer: Interactive formulas, Graphs, Examples Dashboard: Continue learning, Progress, XP, Saved lessons Search System: Global search

MVP Success Criteria: 1000 Users 10 Simulations 100 Lessons 95+ Lighthouse Score

PHASE 2 – VERSION 1.0

Duration: 8 Weeks

Goal: Create complete learning platform.

Features: AI Tutor Virtual Labs Gamification Achievements Bookmarks Learning Paths Daily Challenges Leaderboards Physics Dictionary Study Notes

Success Criteria: 5000 Users 50 Simulations 500 Lessons

PHASE 3 – VERSION 2.0

Duration: 8 Weeks

Goal: Community learning platform.

Features: Community Discussion Forums Study Groups Teacher Dashboard Assignments Certificates Class Management Shared Notes

Success Criteria: 20,000 Users 100 Simulations 1000 Lessons

PHASE 4 – VERSION 3.0

Duration: 12 Weeks

Goal: World-class interactive ecosystem.

Features: Physics Sandbox Experiment Builder Research Hub Competition System AI Recommendations Adaptive Learning Personalized Dashboard Creator Platform

PHASE 5 – FUTURE PLATFORM

Goal: Become global learning ecosystem.

Features: Mobile App Desktop App AR Labs VR Labs Marketplace School Platform Institution Platform API Platform

RELEASE STRATEGY

Alpha: Private Testing Beta: Early Users Public Launch: MVP Release Version 1: Feature Complete Version 2: Community Release Version 3: Immersive Ecosystem

TEAM STRUCTURE

Product Lead Frontend Team Backend Team 3D Team AI Team Content Team QA Team DevOps Team

SPRINT STRUCTURE

2 Week Sprints Sprint Planning Development Testing Review Deployment

TECHNICAL MILESTONES

Milestone 1: Foundation Milestone 2: Learning System Milestone 3: Simulations Milestone 4: AI Tutor Milestone 5: Community Milestone 6: Advanced Labs Milestone 7: Global Launch

MONETIZATION ROADMAP

Phase 1: Free Platform Phase 2: Premium Subscription Phase 3: School Plans Phase 4: Certifications Phase 5: Marketplace Revenue

PRODUCT SUCCESS METRICS

100K Users 500+ Simulations 5000+ Lessons 100+ Virtual Labs Global User Community

FINAL VISION

Become the world's most immersive and intelligent physics learning ecosystem.